# H.D.R.A Report

Hazards and Disasters Risk Assessment

## 1. Financial Impact (Present Value Capitalization)

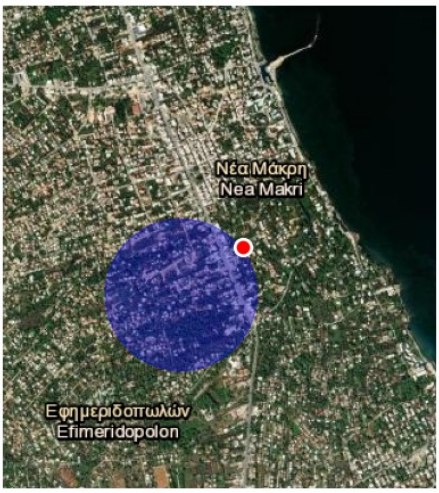
SLE: 5000.0 EUR | ARO (50yr ref): 0.09 | ALE: 450.0 EUR/year

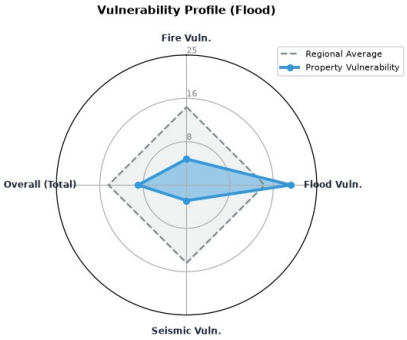
**Final Property Value: 21500.0 EUR (Immediate Market Drop: -14.0%)**

** SLE (Single Loss Expectancy): Expected financial loss from a single disaster event.*
** ARO (Annualized Rate of Occurrence): Estimated annual event frequency based on 50-year historical data.*
** ALE (Annualized Loss Expectancy): Expected annual financial cost of risk (SLE x ARO).*

## 2. Details

| Attribute | Value |
|---|---|
| Scenario | Flood |
| Initial Property Value | 25000.0 EUR |
| Floor Level | Ground |
| Slope (EU-DEM 25m) | 3.4% |
| Construction Material | Concrete |
| Construction Standards | Post-1999 |




## 3. Structural Impact Factors (Hierarchical Weighting)

| Property Characteristic | Calculated Impact / Vulnerability Penalty |
|---|---|
| Material: Concrete | High structural resilience (+0% penalty) |
| Standards: Post-1999 | Modern anti-seismic code (-2% Seismic) |
| Floor: Ground | Moderate Flood vulnerability (+15% Flood) |
| Slope: 3.4% | Optimal drainage & stability (+0% penalty) |
| Roof: Tiles | Minor ember penetration risk (+3% Fire) |
| Vents: None | Sealed against fire embers (-3% Fire) |

## 4. Last Recorded Disaster Metadata (8-Mile Buffer / 50-Year Horizon)

**--- Flood Hazard ---**

Data Source: Copernicus ERA5 / Ministry of Climate Crisis (Live API) (Retrieved: 2026-10-14) | Event Date: 2026-04-01 | Total Events (50yrs): 18
Event Details: Rainfall: 58.5mm in 24h (Peak: 65.4mm). Severe flash flood risk.

*Terms & Privacy (GDPR): This H.D.R.A. report is an automated econometric & geospatial estimation based on Copernicus/EMSC/EFFIS data. It does not substitute a certified property appraisal or structural engineering inspection. Coordinates are processed anonymously.*